



Concepts of : Flow, Activities and Space

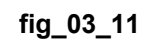
March 2026
W Kruger



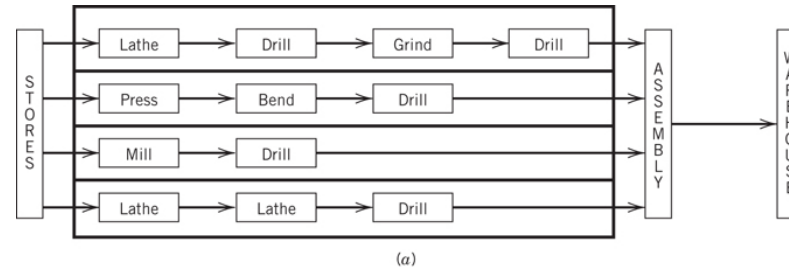
Flow systems in a typical manufacturing facility

- Materials Management system = purchasing
 - Info associated with it include: Production forecasts, demand forecasts, inventory records, purchase orders, stock requisitions, bills of lading, receiving reports, move tickets, order payments
- Material Flow system = production
 - Info associated with it include: Production schedules, work order releases, bar codes, route sheets, assembly charts, warehouse records
- Physical distribution system
 - Info associated with it include: sales orders, packing lists, shipping reports and releases, invoices, bills of lading
- *Parkinson's Law : "Things will expand to fill all available capacity sooner than you plan"*

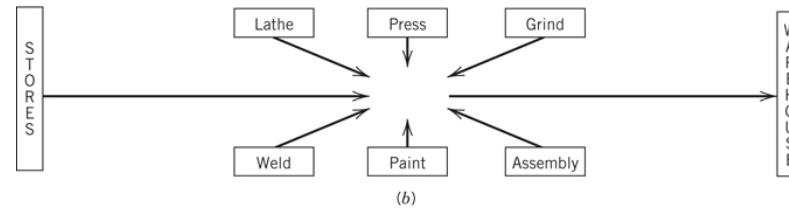
Whether for the whole facility or between 'departments' or areas



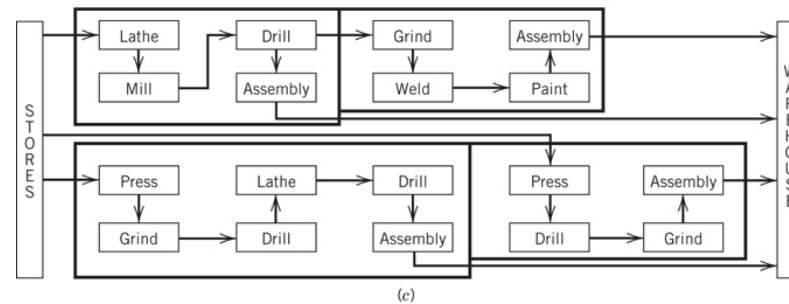
Differences in department definition, influencing layout



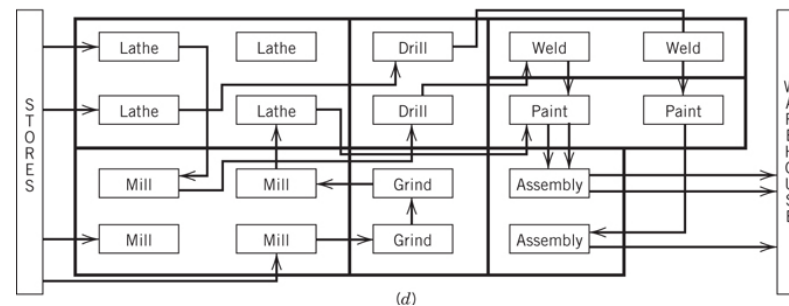
Production line idea



Machine clusters, related per operator?



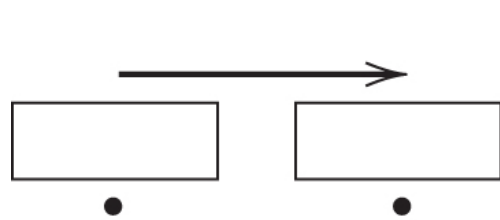
Clusters of machines, related to product clusters



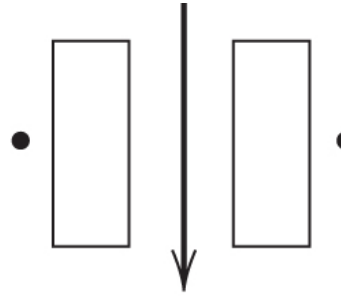
Machine types grouped together

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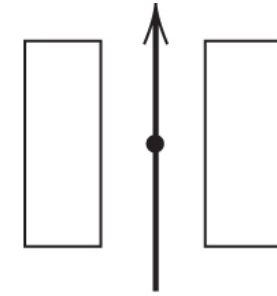
Flow of material within product departments



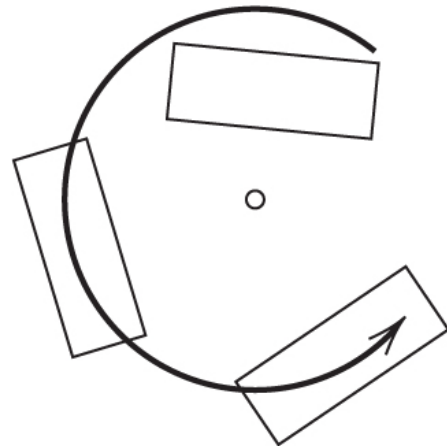
(a)



(b)



(c)



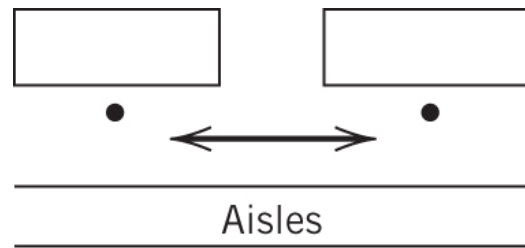
(d)



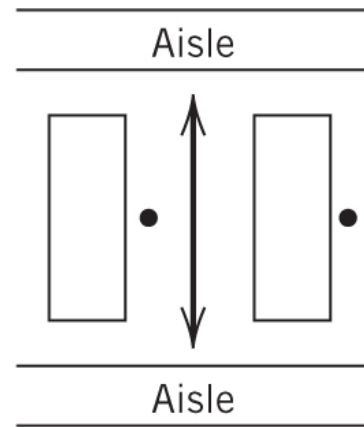
(e)

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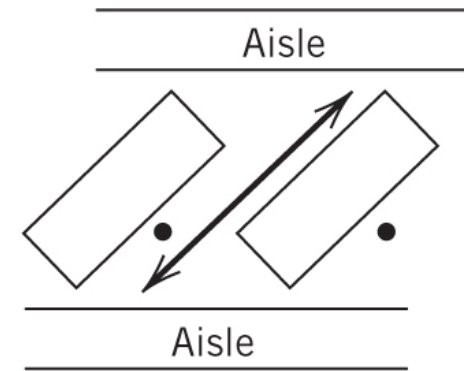
Flow of material within process departments



(a)



(b)



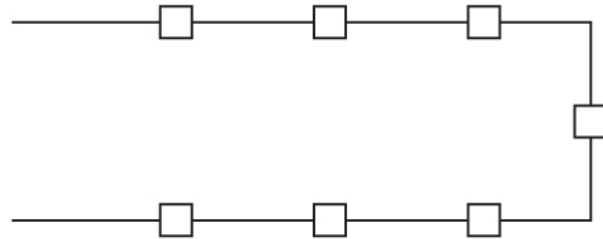
(c)

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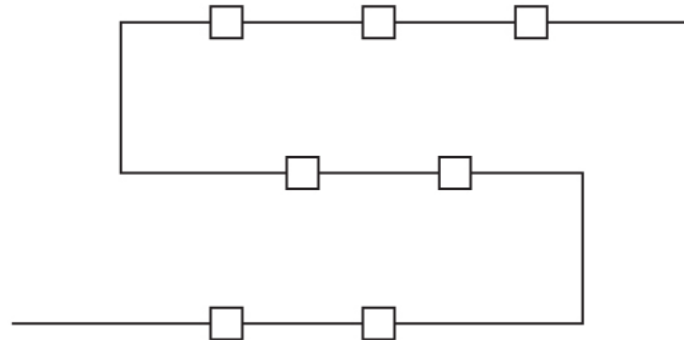
Flow affected by material handling e.g. conveyors, carts, robots, guided vehicles etc



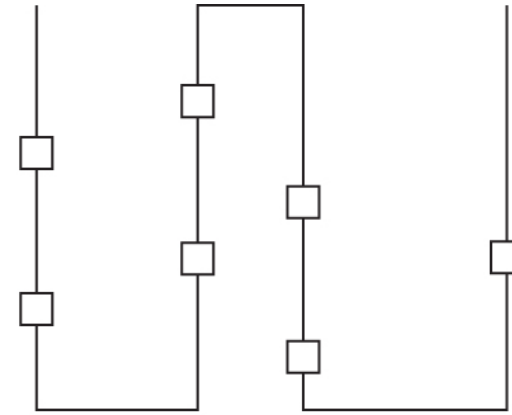
(a) I-flow (straight-line flow)



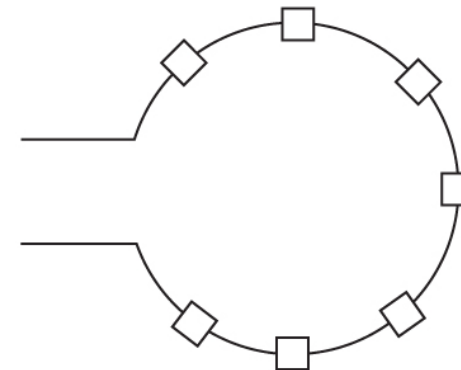
(b) U-flow



(c) S-flow

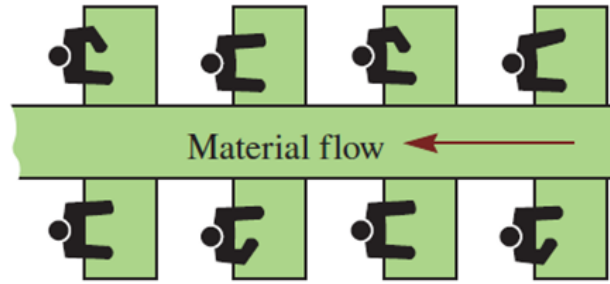


(d) W-flow

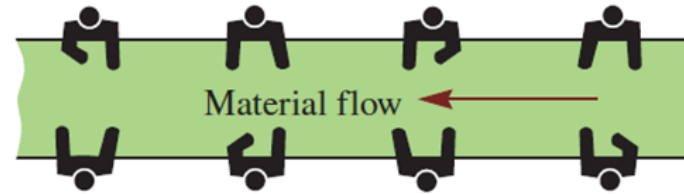


(e) O-flow

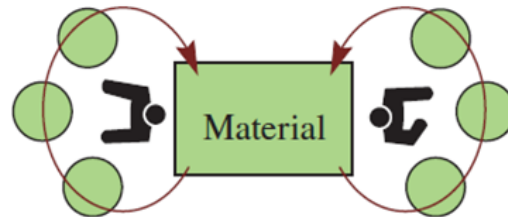
Good vs Bad layouts ito flow



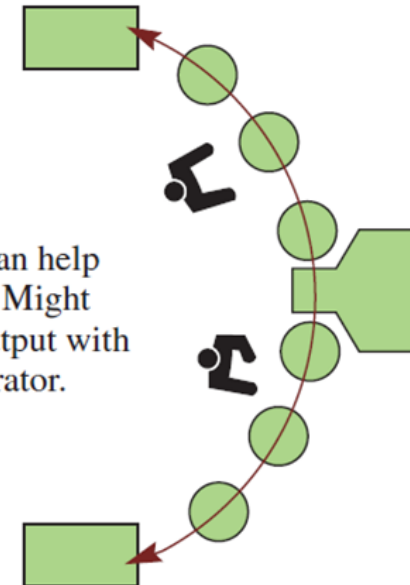
Bad: Operators caged. No chance to trade elements of work between them.
(Subassembly-line layout common in American plants.)



Better: Operators can trade elements of work. Can add and subtract operators. Trained ones can nearly self-balance at different output rates.

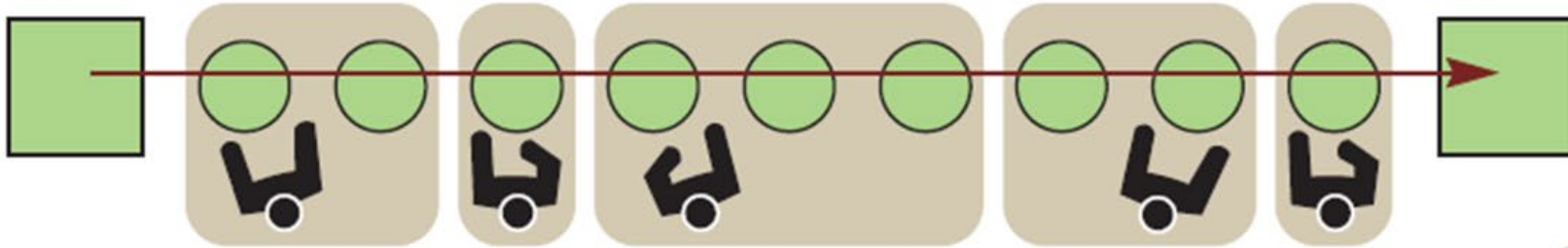


Bad: Operators birdcaged. No chance to increase output with a third operator.



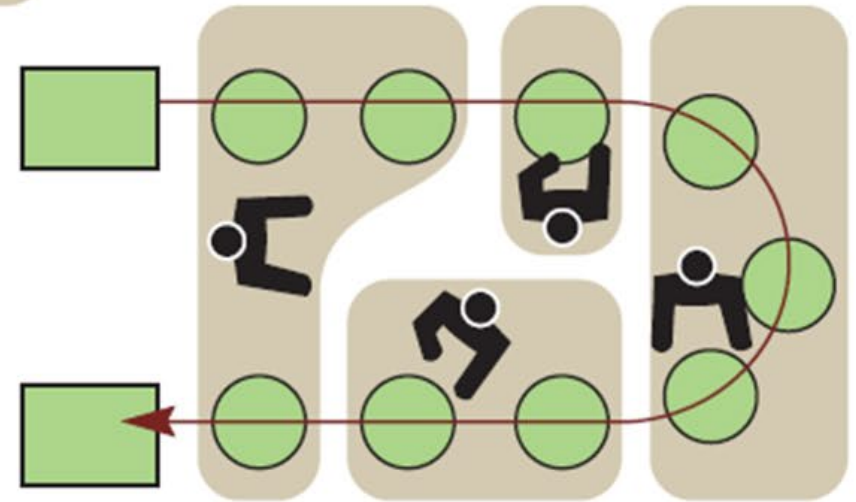
Better: Operators can help each other. Might increase output with a third operator.

Some more₂



Bad: Straight line difficult to balance.

Better: One of several advantages of U-line is better operator access. Here, five operators were reduced to four.



Space specs

Design principle is from Bottom Up

(e.g. from workstation/*desk* to units/*depts* to groups/*faculties* to head office/*campus*)

Space incl. equipment, materials and personnel

Think: workstations, inventory, tools, storage, maintenance, operators, machines

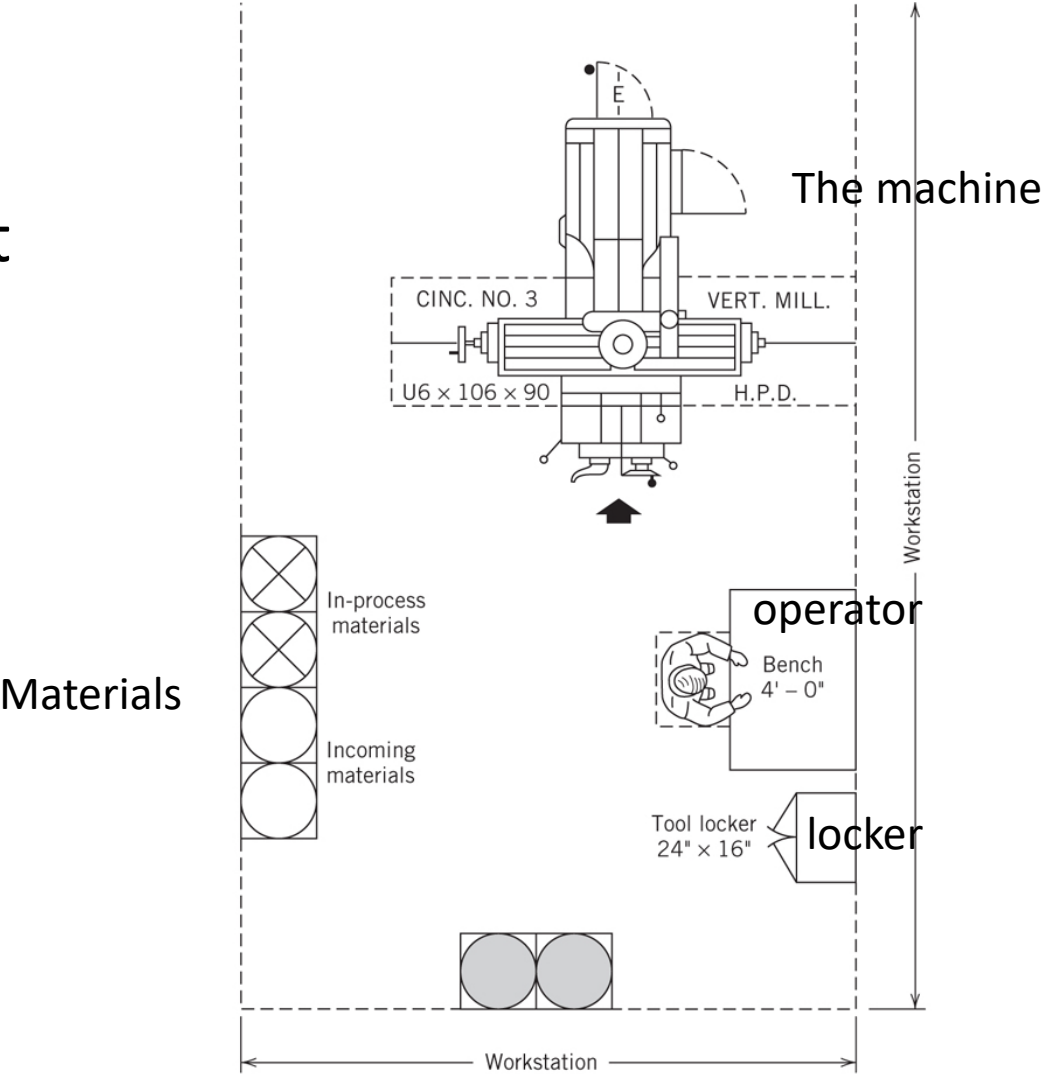
Visualization helps – from above, 2D or 3D

Aisle space to promote *effective* FLOW

- Too narrow will cause congestion
- Too wide is wasted space
- 80-100cm around machines
- Standard specs for people, forklifts, doors, trucks etc.
- Usually calculated as allowance % of netto area

Again, Visualization helps...

This is A
workstation
example....to start
Bottom up design



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Exercise – Calculate floor space (example 3.5)

Data given:

13 machines performing turning operations

5 x Turret lathes (4mx12m) *material storage 20sqm*

6 x Automatic screw machines (4mx15m) *material storage 40sqm*

2 x Chuckers (6mx5m) *material storage 50sqm*

Personnel footprints 4mx5m (1 operator per machine)

Include Supervisor desk 4sqm

Aisle space allowance of 13% (round off)

Total space sqm required = _____

Exercise – Calculate floor space for a department

Equipment	Equipment area needed	Material area needed	Personnel area needed	Total
Lathes (5)				
Screw m/c (6)				
Chuckers (2)				
			Nett total + desk	
			Aisle %	
			TOTAL AREA	

Exercise – Calculate floor space

Equipment	Equipment area needed	Material area needed	Personnel area needed	Total
Lathes (5)	$4 \times 12 = 48 \times 5 = 240$	$20 \times 5 = 100$	$20 \times 5 = 100$	$240 + 100 + 100 = 440$
Screw m/c (6)	$4 \times 15 = 60 \times 6 = 360$	$40 \times 6 = 240$	$20 \times 6 = 120$	$360 + 240 + 120 = 720$
Chuckers (2)	$6 \times 5 = 30 \times 2 = 60$	$50 \times 2 = 100$	$20 \times 2 = 40$	$60 + 100 + 40 = 200$
			Nett total	$440 + 720 + 200 = 1360 + \text{desk} = 1364$
			Aisle %	177
			TOTAL AREA	$1364 + 177 = 1541 \text{sqm}$

Management approaches also influence design e.g. Visual factory scenario below

